

Love Your Heart

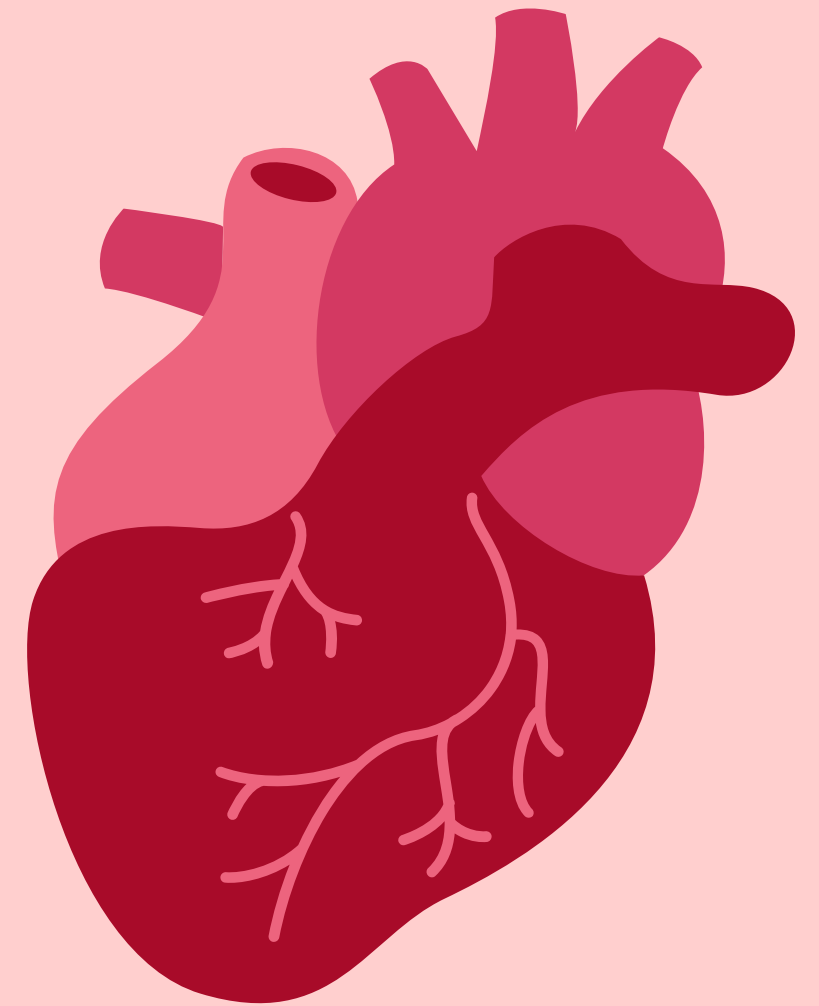
AI powered early Heart Risk Detection

Final Project, DataScience Bootcamp



WBS Coding School

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Why it matters?



Heart disease is a leading cause of death.

1/3

90%

The percentage of Heart disease could be prevented with early detection

From research, the amount could reduce heart risk with control over modifiable factors

55%



Develop a small, accessible application that offers timely, personalized guidance



What We Built

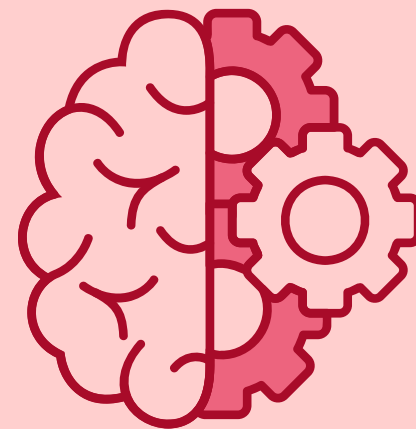
An interactive Streamlit-App with an ML Model to predict heart-risk probability

User-Input



Age, BMI, physical activity, smoking, diabetes, stroke history, etc.

ML Model



Our pipeline processes inputs using a **Voting Classifier** model.

Prediction



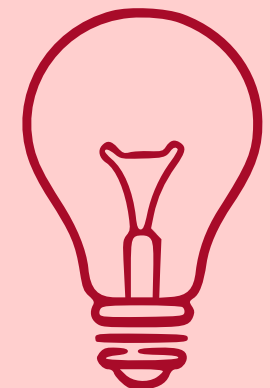
Model generates a probability of belonging to the heart attack class in tiers: **Low / Medium / High**

Importances



We inform the user about the factors that influenced the score.

AI-Suggestions



Lifestyle tips tailored to the user's data.

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♥ Get your heart event score

> How it works

Your Data What if?

Demographics

Select your Sex: ⓘ

Female

Male

Enter your age:

56

⊙ - +

Which of these groups best represents your race?

White

⊙ ▼

> General Health and Lifestyle

> Disease History

Please fill out all fields to enable saving your profile.

💾 Save Profile & Get Score



✓ Low Risk. Maintain a healthy lifestyle to keep your risk low.

What Influenced Your Score

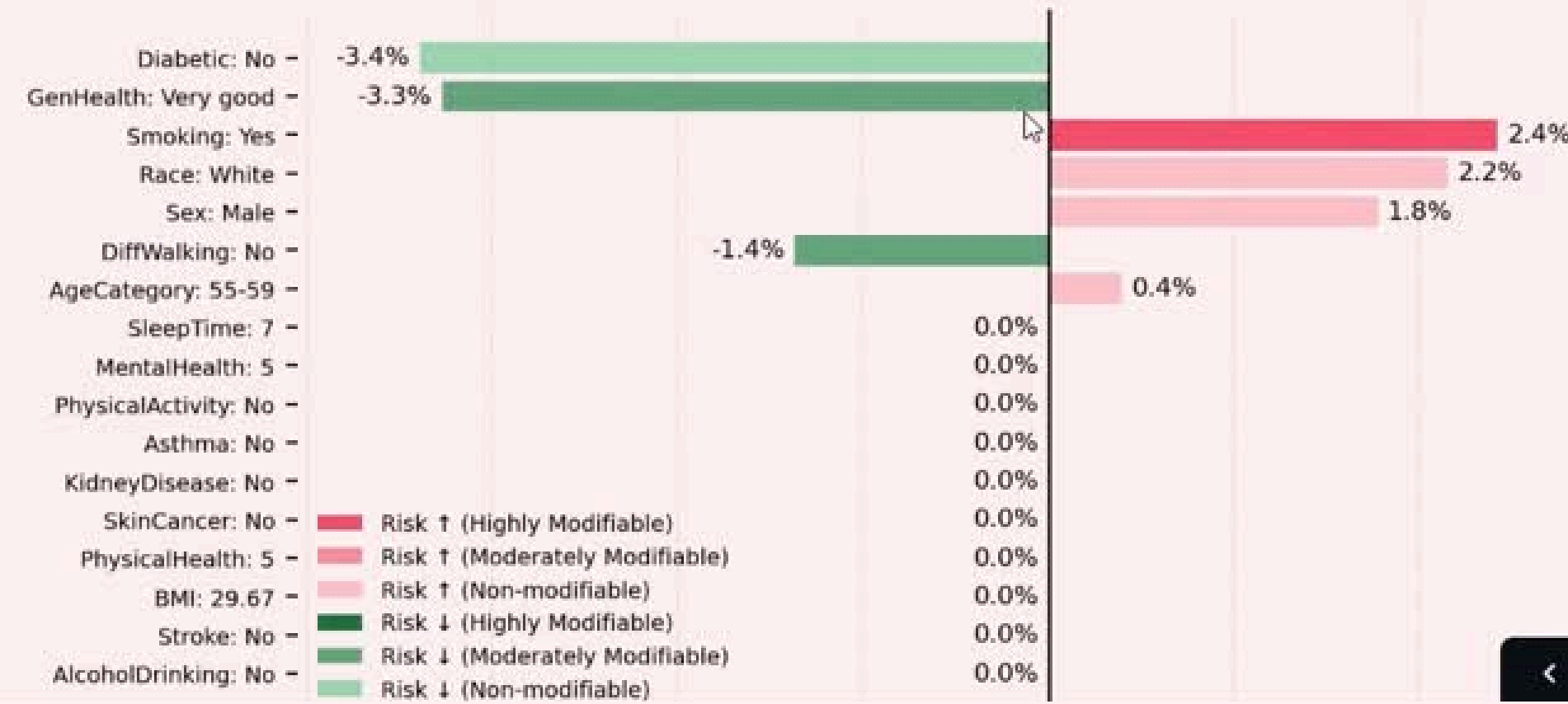
Priority actions: highly modifiable factors

● Smoking: +2.422

Secondary actions: moderately modifiable factors

No moderately modifiable factors currently increasing risk.

> What Are Modifiable Factors?



Do you have trouble walking or climbing stairs?

NoYes

Were you ever a smoker?

NoYes

How many alcoholic drinks do you drink per week?

3-+

Did you exercise in the last 30 days?

NoYes

On how many days in the last 30 days:

Was your physical health not good?

5

Was your mental health poor?

5

> Disease History

Save 'What if'-Scenario & Get Score



✔ Low Risk. Maintain a healthy lifestyle to keep your risk low.

What Influenced Your Score

Priority actions: highly modifiable factors

● Smoking: +2.422

Secondary actions: moderately modifiable factors

No moderately modifiable factors currently increasing risk.

> What Are Modifiable Factors?



Are you satisfied with the results of the assessment?

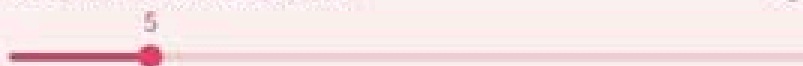
No Yes

On how many days in the last 30 days:

Was your physical health not good?



Was your mental health poor?



> Disease History

Save 'What if'-Scenario & Get Score

'What-If' scenario saved!

Your BMI would be: 29.67

Low Risk. Maintain a healthy lifestyle to keep your risk low.

Updating importances...

What Influenced Your Score

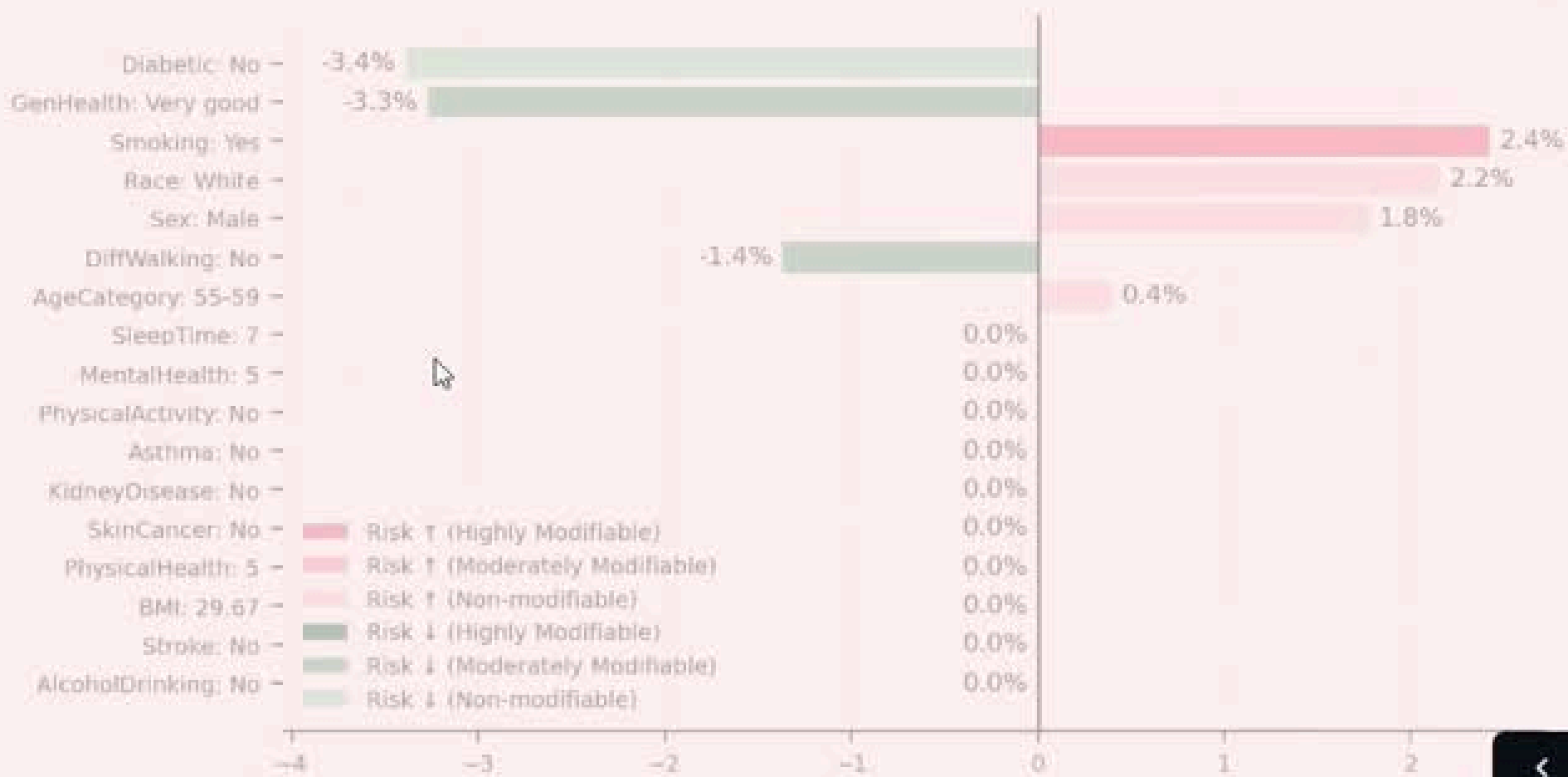
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♥ AI assistant

▼ 📄 Your Profile

Demographic

Sex: Male

Age: 56

Race: White

Risk: 18.11%

General Health

General Health: Very good

Weight: 94 kg

Height: 178 cm

Bmi: 29.67

Sleep Time: 7 hours/day

Physical Activity: No days/month

Physical Health Poor: 5 days/month

Mental Health Poor: 5 days/month

Difficulty Walking: No

Smoking: Yes

Alcohol Consumption: No

Diseases

Diabetic: No

Stroke: No

Asthma: No

Kidney Disease: No

Skin Cancer: No

▼ My recommendation for you:

Ask me anything!

< Manage app

What's Next



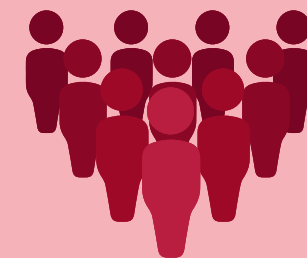
Clinical Features

Incorporate lab results to improve predictive accuracy



Monitoring

Capture trends in vitals to estimate evolving risk



Population Fit

Adjust model to reflect the German population

Thank you!



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Nadine Jacobsen



<https://loveyourheart.streamlit.app/>



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